

Database Design Guide

These project will help to manage or handle the Medicine Inventory and also Tracking the expiry that make easy to track the batches..

roles

.users

.medicine_categories

.medicine

.suppliers

.medicine_batches

.purchase_orders

.purchase_order_items

.sales

.sale_items

.stock_transactions

.expiry_alerts

1. Introduction to MySQL

MySQL is an open-source relational database management system (RDBMS) that uses structured query language (SQL) to manage and manipulate data in a database. It is widely used for various applications, from small web applications to large enterprise systems.

MySQL's key features include:

- **Scalability:** Capable of handling large amounts of data and concurrent connections.
- **Flexibility:** Supports various data types and storage engines.
- **Performance:** Optimized for speed and efficiency.
- **Reliability:** Known for its stability and robustness.

2. Installation of MySQL

MySQL can be installed on various operating systems, including Windows, macOS, and Linux. Here are the general steps to install MySQL:

Windows:

- Download the MySQL installer from the official website.
<https://dev.mysql.com/downloads/installer/>
- Run the installer and follow the on-screen instructions.
- Choose the installation type (Typical, Complete, or Custom). Recommended Custom.
- Set a root password for the MySQL server.

3. E-R Diagram (ERD)

An Entity-Relationship Diagram (ERD) is a visual representation of the data model that shows the entities, attributes, relationships between entities, and cardinality. ERDs are commonly used in database design to help developers and stakeholders understand the structure and relationships within a database.

Identify Entities

- Start by identifying the main entities in your system. These are the objects or concepts about which you want to store data.
- Each entity should correspond to a table in your database.

Identify Relationships

- Determine how entities are related to each other. There are three types of relationships: one-to-one (1:1), one-to-many (1:N), and many-to-many (N:M).
- Represent these relationships using lines connecting the entities.

Refine and Review:

- Review your ERD with stakeholders and team members to ensure it accurately represents the data model and relationships. Make any necessary refinements.

Create the Diagram

- Use specialized diagramming software or tools (e.g., Lucidchart, draw.io, or even pen and paper) to create your ERD.

Optional: Add Attributes and Constraints

- Include additional information in your ERD, such as primary keys, foreign keys, and constraints (e.g., unique constraints).

Table Structure

1. Roles

```
mysql> SELECT * FROM roles;
```

role_id	role_name	description
1	Admin	Full system access
2	Pharmacist	Manages medicines and sales
3	Store Manager	Manages inventory and suppliers
4	Staff	Handles basic sales operations

```
4 rows in set (0.00 sec)
```

2. Users

```
mysql> SELECT * FROM users;
```

user_id	role_id	name	email	password_hash	status	created_at
1	1	Admin user	admin@medicine.com	admin123	Active	2026-08-10 22:46:23
2	2	John pharmacist	john@medicine.com	johnn123	Active	2026-08-10 22:46:23
3	3	Sarah manager	sarah@medicine.com	sarah123	Active	2026-08-10 22:46:23
4	4	Minu staff	minu@medicine.com	minu123	Active	2026-08-10 22:46:23

```
4 rows in set (0.00 sec)
```

3. Medicine_categories

```
mysql> SELECT * FROM medicine_categories;
```

category_id	category_name	description
1	Antibiotics	That medicine used to treat infections
2	Painkillers	That medicine used to reduce pain
3	Vitamins	Vitamins and nutritional supplements
4	Antacids	Medicine used for acidity

```
4 rows in set (0.00 sec)
```

4. Medicine

```
mysql> SELECT * FROM medicine;
```

medicine_id	medicine_code	name	generic_name	category_id	dosage_form	strength	unit	record_level	description	status
1	MED001	Paracetamol	Paracetamol	1	Tablet	500mg	Tablet	20	For fever and pain	Active
2	MED002	Amoxicillin	Amoxicillin	2	Capsule	500mg	Capsule	10	For Infection	Active
3	MED003	Vitamin C	Ascorbic Acid	3	Tablet	500mg	Tablet	15	Vitamin supplement	Active

```
3 rows in set (0.00 sec)
```

5.Suppliers

```
mysql> SELECT * FROM suppliers;
```

supplier_id	supplier_code	name	contact_person	phone	email	address	gst_number	status
1	SUP001	ABc Pharma	raj	9846464634	abc@gmail.com	Mumbai	GST001	Active
2	SUP002	Medicare	Amit	9587126545	medicare@gamil.com	Pune	GST002	Active

```
2 rows in set (0.00 sec)
```

6.Medicine_batches

```
mysql> SELECT * FROM medicine_batches;
```

batch_id	medicine_id	supplier_id	batch_number	manufacture_date	expiry_date	purchase_price	selling_price	quantity_received	quantity_available
100	1	1	BATCH001	2026-01-01	2028-01-01	2.00	5.00	100	
50	2	2	BATCH002	2026-02-01	2028-02-01	5.00	8.00	50	
80	3	3	BATCH003	2026-03-01	2027-12-01	3.00	6.00	80	

```
3 rows in set (0.00 sec)
```

7.Purchase_orders

```
mysql> SELECT * FROM purchase_orders;
```

purchase_order_id	po_number	supplier_id	order_date	expected_date	status	subtotal	tax	discount	total_amount	notes
1	P0001	1	2026-08-10	2026-08-15	Pending	500.00	50.00	10.00	540.00	Medicine order

```
1 row in set (0.00 sec)
```

8.Purchase_order_items

```
mysql> SELECT* FROM purchase_order_items;
```

po_item_id	purchase_order_id	medicine_id	quantity_ordered	unit_price	tax	discount	total_amount
1	1	1	100	2.00	20.00	5.00	215.00
2	1	2	50	5.00	25.00	5.00	270.00

```
2 rows in set (0.00 sec)
```

9.Sales

```
mysql> SELECT * FROM sales;
```

sale_id	invoice_number	customer_name	sale_date	subtotal	tax	discount	total_amount	payment_method	status	created_by
1	INV001	Rahul	2026-08-10 10:00:00	25.00	2.00	0.00	27.00	Cash	Completed	2

1 row in set (0.00 sec)

10.Sale_items

```
mysql> SELECT * FROM sale_items;
```

sale_item_id	sale_id	medicine_id	batch_id	quantity	unit_price	discount	tax	total_amount
1	1	1	1	5	5.00	0.00	2.00	27.00

1 row in set (0.00 sec)

```
mysql> |
```

11.Stock_transactions

```
mysql> SELECT * FROM stock_transactions;
```

transaction_id	medicine_id	batch_id	transaction_type	quantity	reference_type	reference_id	transaction_date	notes	created_by
1	1	1	Purchase	100	Purchase	1	2026-08-10 09:00:00	Stock recieved	2

1 row in set (0.00 sec)

12.Expiry_alerts

```
mysql> SELECT * FROM expiry_alerts;
+-----+-----+-----+-----+-----+
| alert_id | batch_id | alert_type | alert_date | status |
+-----+-----+-----+-----+-----+
|          1 |          1 | 90 Days    | 2026-08-10 | Pending |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Now, let's create the ER diagram to visually represent the entities and relationships.

ERD Diagram..

ERD Diagram – Medicine Inventory System

